

Monitoring devices

Events and alerts

Lenovo

Monitoring functions

XClarity One can monitor the asset inventories, firmware, health status, and event history of managed devices. This information can be accessed by selecting **Monitor** from the context menu. The information is grouped into the following sections:

- **Alerts** are hardware or management conditions that require investigation and user action. XClarity One polls the resource managers and displays alerts that are received from those resources.
- **Events** are all the resource and audit events.
- **Jobs** are long-running tasks that run in the background. A log of all jobs that XClarity One has started is available.
- **Forwarding** can send event data from XClarity One to external applications for monitoring and analysis.



Monitoring alerts

The **Alerts** section of the **Monitor** page contains a donut chart, which shows the total number of critical and warning alerts, and also a bar chart, which shows how old the alerts are.

Click the donut chart to go to the **Alerts** page, where alerts are listed with a description, a severity level, and resource information. By hovering the cursor over the item, the date and time of the alert is also shown.



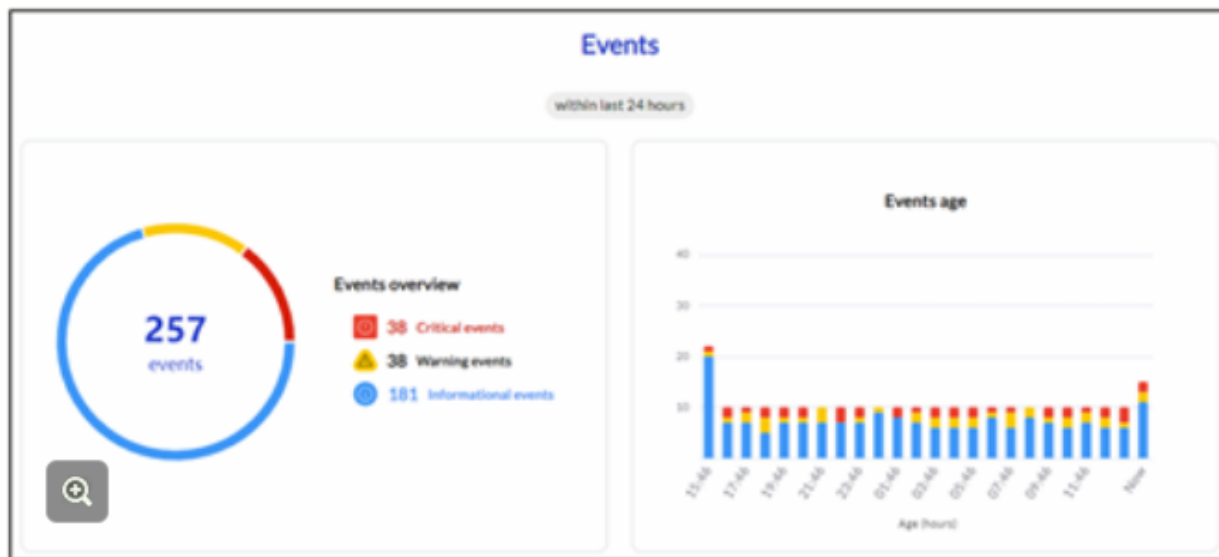
Alert	Severity	Resource	Date
Power supply Power Supply 1 has lost input. When a hardware event is detected by the Lenovo XClarity Controller on the server, the Lenovo XClarity Controller writes that event in the system-event log on the server. Complete the following steps to resolve this problem: 1) Check the LEDs on the PSU. 2) If the AC LED is not lit, check the power cord input and output voltage. 3) If the DC LED is not lit, remove and re-install the power supply. 4) If the problem persists, collect service data and contact Lenovo Support.		Cyborg SR650-31.21 10.241.31.21	Yesterday 04/16/2024 - 16:01
Sensor Phy Presence Set has transitioned from normal to non-critical state.		Cyborg SR650-31.21 10.241.31.21	Yesterday
Device Phy Presence Jmp has been added.		Cyborg SR650-31.21 10.241.31.21	Yesterday

Monitoring events

The **Events** section of the **Monitor** page contains a donut chart and a bar chart. The donut chart shows the total number of critical, warning, and informational events. The bar chart shows how old the events are.

All the information shown here relates to events created in the last 24 hours.

Click the donut chart to go to the **Events** page, where events are listed with a description, a severity level, and resource information. By hovering the cursor over the item, the date and time of the event is also shown.

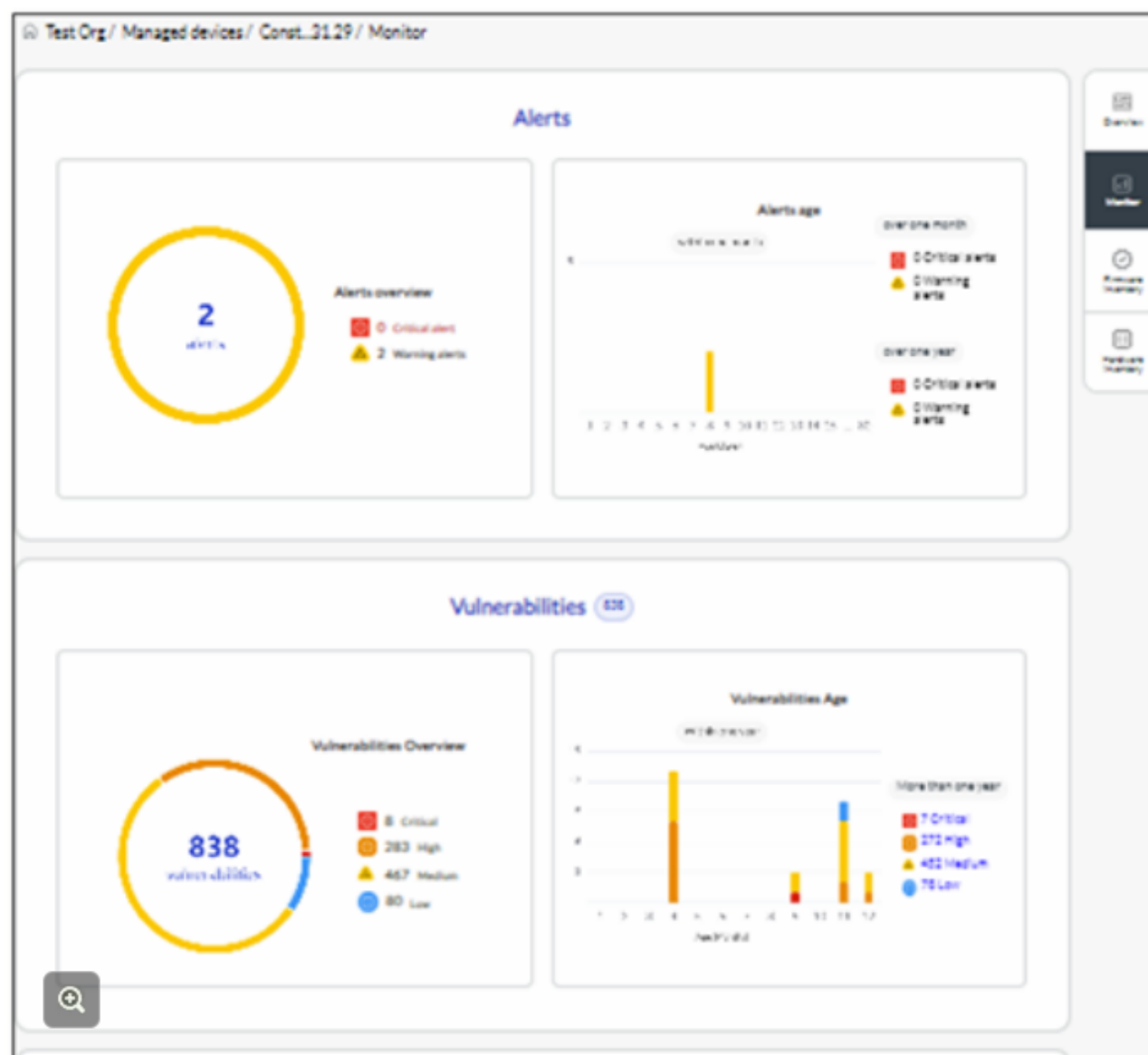


The screenshot shows the 'Events' section with a search bar and a table of events. The table has columns for Event, Severity, Resource, Date, and User. The first three events are informational, and the last one is a warning.

Event	Severity	Resource	Date	User
User npopa@lenovo.com from remote IP address 192.168.6.191 successfully created the collection All servers with 2 members.	Informational	All servers	34 minute(s) ago	npopa@lenovo.com
User npopa@lenovo.com from remote IP address 192.168.6.191 successfully created the collection 7X08 servers with 1 members.	Informational	7X08 servers	34 minute(s) ago	npopa@lenovo.com
User npopa@lenovo.com from remote IP address 192.168.6.191 successfully created the collection 7X04 servers with 1 members.	Informational	7X04 servers	34 minute(s) ago	npopa@lenovo.com
The warranty for device Carnage-SR550-31.23 has expired.	Warning	Carnage-SR550-31.23 10.241.31.23	53 minute(s) ago	-

Monitoring specific device events

Specific devices can be monitored by selecting **Monitor** → **Managed devices** → and then choosing a device. Information about the device will be shown on a new page.



Custom alerts

Setting up custom alerts to monitor a device

Lenovo

XC1 custom alerts

XClarity One can set up thresholds on devices to monitor processor and memory usage. On the settings screen, select **Thresholds** from the context menu on the right.

The following configurable values are available:

- Usage: Select warning and critical thresholds
- Time: Select continuous or a specific period

If the specified threshold is exceeded, an alert will be raised along with a list of actions to take. For example, if memory usage exceeds 52% for a cumulative 10 minutes within a 60-minute window, a warning alert will be triggered.

Custom Alerts Test / Thresholds

Usage Thresholds

Name	Warning threshold	Critical threshold	Alert duration	Details
Memory	52%	54%	10 of 60 minutes	Memory Utilization thresholds for warning and critical alert and alert duration
Processor	52%	54%	10 of 60 minutes	Processor Utilization thresholds for warning and critical alert and alert duration

Context menu on the right:

- Coming soon
- General
- Call home
- Data forwarding
- Thresholds**

Device J100CMM4 has CPU Utilization values above 52 % for 10 out of 60 minutes See details		J100CMM4 10.240.26.34	13 hour(s) ago
Device J100CMM4 has CPU Utilization values above 54 % for 10 out of 60 minutes See details		J100CMM4 10.240.26.34	13 hour(s) ago
Device J100CMM4 has Memory Utilization values above 52 % for 10 out of 60 minutes See details		J100CMM4 10.240.26.34	13 hour(s) ago
 Device J100CMM4 has Memory Utilization values above 54 % for 10 out of 60 minutes See details		J100CMM4 10.240.26.34	13 hour(s) ago

Forwarding monitoring data

Forwarding data by web service or email

Lenovo

Forwarding events

XClarity One can forward data about events to external services for monitoring and analysis. Filters can be used to make sure that only data that meets specific criteria is forwarded. After being set up, data forwarders can be enabled, edited, or deleted.

The following methods can be used to forward data:

- Email: Event data is forwarded to selected email addresses
- Web service: Event data is forwarded over the network to a REST web service

GARegression / Forwarding

Data forwarding 2

Search for anything

☐	Name	State	Description	Type	Destination
☒	rest01	⊖		REST Forwarder	https://dev207895.service-now.com/api/x_757322_lenovo_xc/forwarder_del
☐	Test SMTP	⊖	test	SMTP (Email) Forwarder	psaxena1+gaqa2user@lenovo.com

1 selected

Items displayed 10 1 - 2 of 2

Enable data forwarder

Edit data forwarder

Delete data forwarder

Creating a data forwarder

Work through the following steps to create a data forwarder:

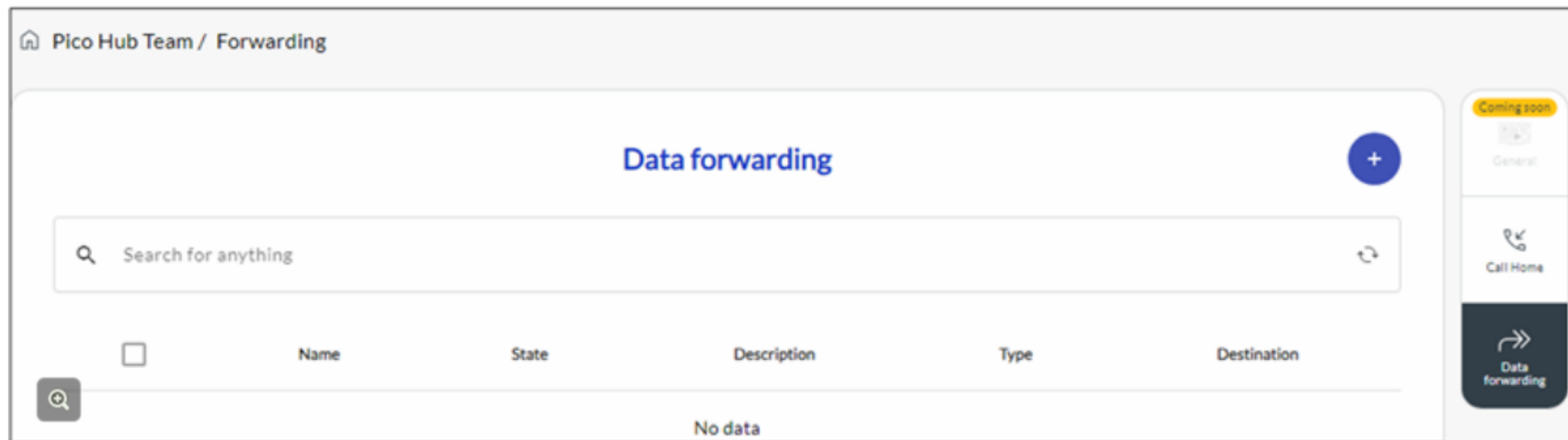
Click each number in turn to see the procedure.

Step



Creating a data forwarder

Step 1. Select **Data forwarding** from the **Settings** menu, and then click  to create a new data forwarder.



Step



Creating a data forwarder

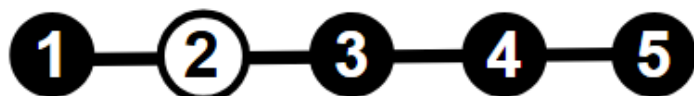
Step 2. The **Create a data forwarder** pop-up window will be displayed. Users can select either a **Web service** or **Email** data forwarder, and then enter a name and description before continuing to the next step.

The image displays two screenshots of the 'Create a data forwarder' pop-up window. Both screenshots show a sidebar on the left with three options: 'Data forwarder' (selected with a green checkmark), 'Configure', and 'Filter'. The main area is titled 'Create a data forwarder' and contains the instruction 'Select the platform where you want to forward data.'.

In the left screenshot, the 'Type*' dropdown is set to 'Web service' (highlighted with a red box). The 'Name*' field contains 'Test Webservice' and the 'Description' field contains 'Test webservice forwarder'.

In the right screenshot, the 'Type*' dropdown is set to 'Email' (highlighted with a red box). The 'Name*' field contains 'Test SMTP' and the 'Description' field contains 'Test SMTP Forwarder'.

Step



Creating a data forwarder

Step 3. Configure the forwarder destination. Click the buttons for more information.

Web service

Email

Configuring the forwarder destination through the web service

- Event data is forwarded over the network to an external REST web service using the HTTPS protocol. Secure port 443 is used by default.
- The **Test Connection** button can be used to test the settings. This test involves XClarity One importing and validating the certificate chain from the web service and then sending a sample event to the web service.



The screenshot shows a configuration window titled "Configure Connection to Web Service". On the left, under "Forwarder", the "Data forwarder" option is selected with a green checkmark, while "Configure" and "Filter" are unselected. The main area contains the following fields and options:

- URL:** A text box containing "https://dev207895.service-now.com/api/x_757322_lenovo_xc/forv".
- Timeout:** A text box containing "30" followed by "Sec".
- Request Method:** Radio buttons for "POST" (selected) and "PUT".
- Authentication Type:** Radio buttons for "User credentials" (selected) and "Token Header".
- Username:** A text box containing "qarestuser".
- Password:** A text box containing masked characters "*****" with a toggle icon on the right.
- Test Connection:** A button at the bottom.

A magnifying glass icon is visible in the bottom left corner of the dialog box.

Step



Creating a data forwarder

Step 3. Configure the forwarder destination. Click the buttons for more information.

Configuring the forwarder destination through SMTP for an email data forwarder

- Users can forward event data to email addresses for one or more users in the organization.
- Users cannot send data to external users.
- The emails are sent from noreply@xclarityone.lenovo.com with an **Event forwarding** subject.

Web service

Email

The screenshot shows a web interface for configuring an email forwarder. On the left, a sidebar titled 'Forwarder' has three options: 'Data forwarder' (selected with a green checkmark), 'Configure' (with a blue circle), and 'Filter' (with a grey circle). The main area is titled 'Configure Email' and contains the following fields:

- From:** A text field containing 'noreply@xclarityone.lenovo.com'.
- Recipients:** A dropdown menu showing a list of email addresses. The visible list includes:
 - ☒ hdhanabalan+qatest1@lenovo.com
 - ☒ hsharma+qa1@lenovo.com
 - ☐ lbirzu@lenovo.com
 - ☒ kmaharana+test1@lenovo.com
 - ☐ mdragomir+1@lenovo.com

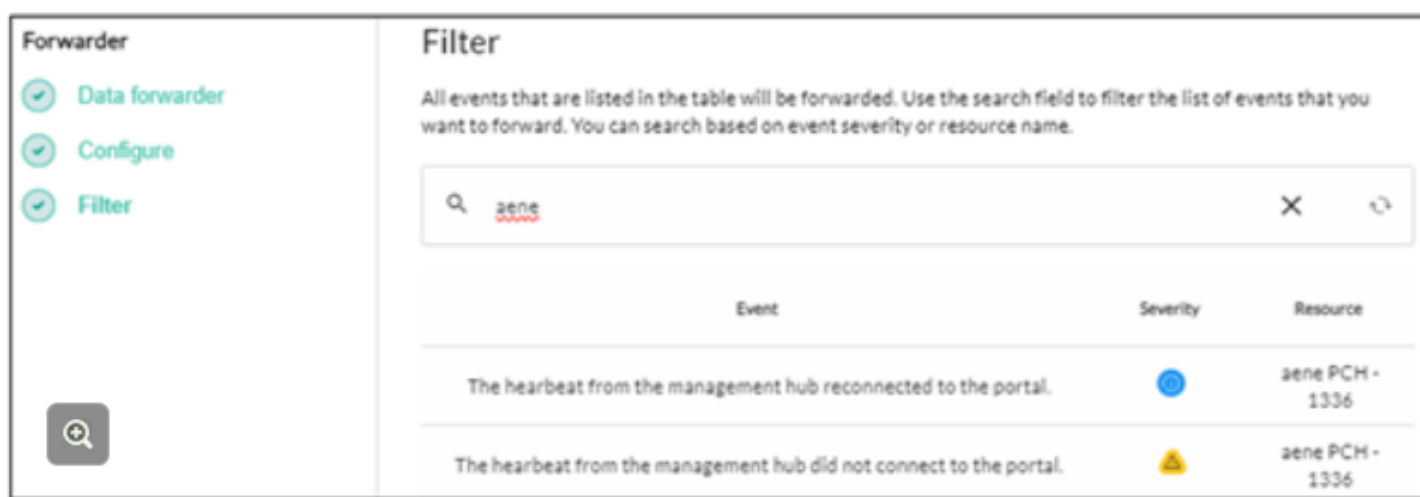
Step



Creating a data forwarder

Step 4. Configure the event filter

- Filters can be used so that not all data is forwarded. Events can be selected based on the resource or device component that generated the event, or by event severity (informational, warning, critical).
- If you do not provide criteria, data is forwarded for all events generated by all resources.
- Users have the option to enable/disable the data forwarder.



Step



Forwarding monitoring data

Creating a data forwarder

Step 5.

An example of an event forwarded by email

[External] Event Forwarding

XClarityOne <noreply@xclarityone.lenovo.com>
To: kmaharana@test1@lenovo.com, prajneesh@test1@lenovo.com, psasena1+qatest1@lenovo.com

action_id	eventCode	service	severity	eventClass	source	targetResource	manager	fruType	messageArgs	_links
new	QXXXB0000271	none	informational	audit	test_connection	- id: DUMMY075E90B40E48CDA2C136AAC7AF9 - type: XClarity Management Hub - name: TEST_Hub	- id: DUMMY075E90B40E48CDA2C136AAC7AF9 - type: XClarity Management Hub - name: TEST_Hub - ipAddress:	other	noreply@xclarityone.lenovo.com	- rel: self - uri: /api/v1/monitoring/events/510

An example of an event forwarded by the web service to an external application

servicenow

Log

Created on Today

Created	Level	Message
2024-04-10 13:34:42	Error	Error accessing descriptor for metaObject: package_private:script:include: function: Login: no thrown error
2024-04-10 13:34:42	Error	Error accessing descriptor for metaObject: package_private:script:include: function: Login: no thrown error
2024-04-10 13:34:42	Information	POST data coming from ATLAS {"message":{"action":"new","id":"12345","category":"other","eventCode":"FQ0XQ0M0002","service":"none","severity":"informational","eventClass":"audit","source":{"id":"7890","type":"AAAAAF7F1ABACCEB31F10502DAC9B1","type":"DeviceGroup","name":"Puma collection"},"manager":{"type":"other","messageArgs":{"psasena1+qatest1@lenovo.com":"104.232.236.6"},"Puma collection":{"target":"A91924CF1F9F42A095AC34ED878FF3","accountId":"7509A18-18014","remoteAddress":"104.232.236.6","remoteCaller":"xclp-groups","username":"psasena1+qatest1@lenovo.com","userId":"104.232.236.6","language":"en","requestType":"POST","requestUri":"/api/v1/monitoring/events","requestPath":"/api/v1/monitoring/events/510"}}

Step



Monitoring jobs

Jobs details, sub-jobs, and logs

Lenovo

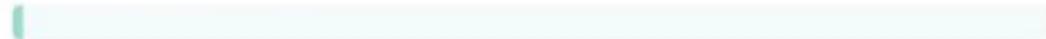
Jobs overview

- A job is a process created by a system or user action. The following actions generate jobs:
 - Unmanage hub
 - Discover device
 - Manage device
 - Unmanage device
 - Power operations
 - Firmware update
 - Create a service ticket
- When a job is started, a pop-up window with a progress bar will be displayed.
- The **Jobs** panel on the **Monitor** page shows how many jobs are in progress.
- Clicking the panel title or the **See jobs** button will redirect the user to the **Jobs** page.

Discovering manageable devices

A job was created to complete this task. If you close this dialog, you can monitor the job status from the Jobs panel, which is accessed from the Monitor tab.

Created: 04/10/2024 - 17:01:33



 Running

See job details

Close

Jobs

There are (2) running jobs.

When a job is started, you can view all the logs for that job.

See jobs

Jobs page

- On the **Jobs** page, root jobs are listed with their title, status, start and (if applicable) end time, and the resource for which the job was generated. By hovering the cursor over a running job, a percentage completion figure can also be seen.
- A maximum of 500 jobs can be listed. If more jobs are created beyond this number, the oldest jobs will be automatically deleted.
- Jobs can also be deleted manually.
- More information about a job can be seen by clicking the job title.



Filtering jobs by device

Starting from a device page, users can navigate to the **Monitor** page and then the **Jobs** page to see only the jobs for that specific device.

- Click on a job to open its sub-jobs, which provide additional details about progress and completion status
- Clicking any sub-job opens a dialog that displays a history of logs related to the device. (Click [HERE](#) for details)

